Q1 [10 Points] John and Sarah have gone together to buy a pack of trading cards. These trading cards each have a value on the trading market, given in the following table.

Card	1	2	3	4	5	6
Value ($)	4	15	20	9	13	7

John and Sarah need to divide up the trading cards so that they each get 3 cards, and also the absolute value of the difference between the value of the trading cards they get is as small as possible. Write down an integer linear program that models this situation.

Solution: Make a variable x_i for $i = 1$ through 6 that is supposed to be 1 if John gets card i and 0 if Sarah gets it. For simplicity, we will let y_i be a variable that is $1 - x_i$ (so that it is 1 if Sarah gets card i, and 0 if John gets card i). If v_i is the value of card i, we want to minimize

$$|\sum_{i=1}^{10} v_i x_i - \sum_{i=1}^{10} v_i y_i| = |\sum_{i=1}^{10} v_i (x_i - y_i)|.$$

To model this with linear constraints, we will let t be a variable representing this absolute value, and introduce the constraints that

$$t \geq \sum_{i=1}^{10} v_i (x_i - y_i),$$

and

$$t \geq -\sum_{i=1}^{10} v_i (x_i - y_i).$$

Combining this with the constraint that the sum of the x_i is 3 yields

$$\begin{aligned} \min \quad & t \\ \text{s.t.} \quad & \sum_{i=1}^{10} x_i = 3 \\ & t \geq \sum_{i=1}^{10} v_i (x_i - y_i) \\ & t \geq -\sum_{i=1}^{10} v_i (x_i - y_i). \end{aligned}$$

Q2 [5 Points]

(a) Here is a linear program

$$\begin{aligned} \text{minimize} \quad & 4x_1 + x_2 + 2x_3 \\ \text{subject to} \quad & 2x_1 + x_2 + x_3 \leq 9, \\ & 3x_1 - x_2 + 2x_3 \geq 4, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Write an equivalent linear program in standard equality form.

Solution: To put a LP in standard equality form, make the objective maximization by negating the objective. Then, introduce slack variables $s_1, \ldots, s_4$ for each inequality, and use this to represent the difference between the inequality and an equality.

$$\begin{aligned} \text{maximize} \quad & -4x_1 - x_2 - 2x_3 \\ \text{subject to} \quad & 2x_1 + x_2 + x_3 + s_1 = 9, \\ & 3x_1 - x_2 + 2x_3 - s_2 = 4, \\ & x_1, x_2, x_3, s_1, s_2 \geq 0. \end{aligned}$$

(b) Consider the following matrix (where the top row just numbers the columns).

$$A = \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix}.$$

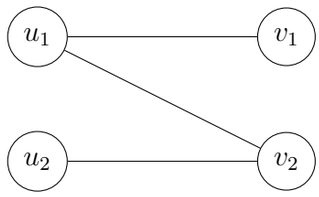
which of the following sets of columns are bases of this matrix?

$$\{2,3,4\}, \{1,2\}, \{3,4\}, \{2,3\}$$

Solution: A basis is a set of linearly independent columns with the same number of elements as there are rows of the matrix.

Only $\{1,2\}$ and $\{2,3\}$ are bases as these columns are linearly independent. $\{2,3,4\}$ has too many elements, and $\{3,4\}$ is linearly dependent.

Q3 [10 Points] Here is a graph.

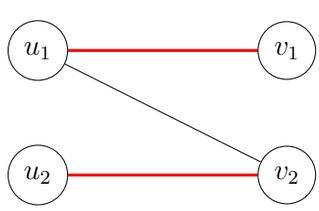


(a) Write down a linear program that computes a maximum matching of this graph. Remember that a maximum matching is a set of edges in the graph where every vertex is incident to at most one edge.

Solution:

$$\begin{aligned} \text{maximize} \quad & x_{11} + x_{12} + x_{22} \\ \text{subject to} \quad & x_{11} + x_{12} \leq 1, \\ & x_{22} \leq 1, \\ & x_{11} \leq 1, \\ & x_{12} + x_{22} \leq 1, \\ & x_{11}, x_{12}, x_{22} \geq 0. \end{aligned}$$

(b) Find an optimal point for the LP written in (a) and justify why it is optimal for the linear program. Hint: a maximum matching for this graph is depicted in the following image. Also, it may be easier to justify why the solution is optimal without writing down a certificate.



Solution: For the linear program above, the optimal solution would set $(x_{11}, x_{12}, x_{22}) = (1, 0, 1)$. To see that this is optimal, note that the objective value is 2, and that summing the inequalities $x_{11} + x_{12} \leq 1$ with $x_{22} \leq 1$ gives

$$x_{11} + x_{12} + x_{22} \leq 2.$$

Q4 [10 Points] For the following linear program, determine whether it is **infeasible**, **unbounded**, or if it has an **optimal point**. Justify your answer.

$$\begin{aligned} \min_{x_1,x_2} \quad & -x_1 + x_2 \\ \text{s.t.} \quad & -x_1 + x_2 \leq 1, \\ & -x_2 \leq 0, \\ & x_2 \leq 1. \end{aligned}$$

Solution: This is a linear program of the form

$$\begin{aligned} \min \quad & c^\intercal x \\ \text{s.t.} \quad & Ax \leq b, \end{aligned}$$

where

$$c = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \qquad A = \begin{pmatrix} -1 & 1 \\ 0 & -1 \\ 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

The point

$$x^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

is feasible. Consider the direction

$$d = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

We have

$$Ad = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \leq 0,$$

so d lies in the recession cone of the feasible region. Moreover,

$$c^\top d = -1 < 0.$$

Hence, for every $t \geq 0$,

$$x^{(0)} + td = \begin{pmatrix} t \\ 0 \end{pmatrix}$$

is feasible, while

$$c^\top (x^{(0)} + td) = -t \longrightarrow -\infty.$$

Therefore the linear program is unbounded below.

End of quiz